Chest X-Ray Classifier Project Report

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Introduction:

Pulmonary diseases such as COVID-19, tuberculosis, and pneumonia remain the leading cause of mortality worldwide. Chest X-rays are one of the most widely used tools to diagnose these conditions as they are cost-effective, but their interpretation requires significant clinical expertise from doctors such as pulmonologists and respirologists. However, Convolutional Neural Networks (CNNs) have the potential to solve this issue; by automating the analysis of medical images such as X-rays. Research has shown that CNNs and pre-trained learning deep learning models can achieve high classification accuracy for lung diseases from chest X-ray images, helping to accelerate the diagnostic process (Ullah et al. 2023). As a result, CNN-based models have been widely established as the "upcoming" and "leading" approach for automated detection and classification of findings in chest X-ray images (Hasan et al., 2023).

This project develops a CNN-based classifier using Tensorflow and Keras to classify chest X-ray images into six pulmonary disease categories: COVID-19, Emphysema, Normal, Pneumonia-Bacterial, Pneumonia-Viral, and Tuberculosis. The model was built, trained, and then evaluated on a held-out test set, with performance assessed using accuracy, precision, recall, and F1-score.

Dataset:

The ChestX6 Multi-Class X-ray dataset was sourced from Kaggle (Asak, 2024). It contains 18 036 X-ray images across the six classes, as they were pre-resized to 224 by 224 pixels. The dataset is also pre-split into three directories: training (14 551 images), validation (1748 images), and test (1737 images), with each directory containing one subfolder for each class. The six classes represent distinct pulmonary conditions. Normal images show healthy chest X-rays, the remaining classes each illustrate unique visual indicators of disease – from fluid in the lungs (Pneumonia-Bacterial) to foggy/unclear lungs (COVID-19).

The class distribution across the training set is relatively balanced, having Normal (2,671), Pneumonia-Bacterial (2,400), Pneumonia-Viral (2,413), COVID-19 (2,417), Tuberculosis (2,600), and Emphysema (2,050) images. Having this near-balanced distribution is a significant advantage since it reduces the risk of the model developing a bias towards a single class during training.

Contributions:

The layout, organization, and evaluation strategy were designed and implemented independently using TensorFlow and Keras. This included the strategy of using a four-block convolutional architecture with batch normalization and dropout and choosing and setting up training callbacks to assist and help the model during training. The graphs, classification report and confusion matrix were all developed to help illustrate and see better insights into how the model performed. In all, TensorFlow and Keras were used for model building and training, Numpy for calculation operations, Matplotlib was used for visualization and scikit-learn was used for evaluation metrics.

Claude was used as an AI assistant during the development of the project. Specifically, Claude assisted with loading the dataset into Google Collab, by suggesting the approach of uploading the dataset as a zip file and extracting it directly within the notebook, as the dataset size posed significant compatibility challenges with Google Collab file system (it could not handle the large file size). Claude also assisted with debugging, when errors were encountered during coding that couldn't be solved independently.

Creativity and Complexity:

Multi-class classification of lung diseases from chest X-rays is considered technically challenging, as distinguishing between conditions can be difficult since some conditions appear to be visually similar on chest X-rays – specifically Pneumonia-Bacterial, and Pneumonia-Viral. Since several pulmonary conditions share overlapping visual characteristics, trained radiologists can find it difficult to classify which condition is which (Hasan et al., 2024).

The model was designed by using a four-block CNN architecture, as each convolutional block increases in filter depth (from 32, to 64, to 128 and 256). The model first learns simple patterns but progressively learns and extracts more complex patterns and details which will help classify the images. Batch normalization was applied after each convolutional layer to stabilize training, and dropout was incorporated in the fully connected layers to reduce overfitting. Data augmentation methods such as random rotation, horizontal flipping, shifting, and zooming were used on the training data to help the model perform better on new data.

Three training callbacks were used to improve training. Early stopping was used to stop training when validation loss no longer improved, model checkpointing to save the best version of the model and learning rate reduction to adjust the optimizer when progress slowed.

Performance Evaluation:

The model was evaluated on the test set of 1737 images across the six classes, as it achieved an overall accuracy of 74% with a macro-average F1-score of 0.73. Here shows each class's precision, recall and F1-score:

Class	**Precision**	**Recall**	**F1-Score**
Covid-19	0.74	0.72	0.73
Emphysema	0.69	0.72	0.71
Normal	0.71	0.98	0.82
Pneumonia-Bacterial	0.70	0.50	0.58
Pneumonia-Viral	0.59	0.56	0.58
Tuberculosis	1.00	0.94	0.97

The model performed strongest on Tuberculosis (F1: 0.97) and Normal (F1: 0.82), demonstrating that these conditions have enough distinct visual patterns for the model to learn effectively, within 10 epochs. The model performed the weakest on Pneumonia-Bacterial and Pneumonia-Viral (F1: 0.58 each). The confusion matrix reveals that Pneumonia-Bacterial cases were frequently misclassified as Pneumonia-Viral (106 out of 300), and Pneumonia-Viral cases were frequently misclassified as Normal (67 out of 300). This is expected, as both bacterial and viral pneumonia share similar radiological features that are hard to distinguish and classify correctly (Hasan et al., 2024).

The training accuracy steadily increased from 47% in the first epoch to 72% by the tenth epoch, showing that the model was learning over time. However, the validation accuracy was quite unstable, ranging between 17% and 71% across different epochs. This fluctuation is likely caused by the relatively small validation set of 1748 images across six classes, which makes the results more sensitive to random changes from one epoch to another. Despite this variability, the model still performs reasonably well on the test set, indicating that it has learned useful patterns and details and can classify effectively to new data. By giving more training time and add additional epochs, the model would likely continue to improve, potentially achieving higher accuracy and more stable validation results.

Ethics:

Using AI in healthcare raises important ethical concerns, that need to be addressed carefully. The most immediate risk is misclassification, since a model that incorrectly identifies a disease class could lead to delays in treatment, and even wrong treatment to the patient. This relates to this model, as based off of the results, the model sometimes confuses Pneumonia-Bacterial and Pneumonia-Viral, with 106 out of 300 bacterial cases being classified incorrectly. Also bias in medical AI can also lead to poor clinical decisions and may worsen existing healthcare inequalities (Cross et al., 2024). As a result, and due to these major risks, the model is not fit enough to be used in a clinical or diagnostic context.

A major ethical concern in medical AI is dataset bias. The ChestX6 dataset does not include important details like patient age, sex, or location. As a result, it is unclear whether the model works equally well for different groups of people. There have been studies that have shown that AI models using chest X-rays can make more mistakes, such as missing diseases for certain racial or demographic groups (Driessen et al., 2024). A study published in Nature Medicine further confirms that these models sometimes rely on hidden demographic patterns when making predictions, and fixing this can actually reduce performance on new data (Obermeyer et al., 2024). This illustrates why it is important to use diverse and representative data when building fair and reliable medical AI systems.

From an equity and inclusion perspective, access to AI-based diagnostic tools is not the same everywhere. People in low-income or under-resourced areas may not have the technology or infrastructure needed to use these tools, and more importantly, they are often not well represented in the data used to train them. Therefore, designing AI systems with fairness and ethics in mind is essential from an engineering perspective (Mithila, 2026). Similarly, Google's responsible AI principles emphasize in reducing bias, and ensuring that AI systems are socially beneficial, which is especially important when relating to healthcare (Google, 2018).

Lastly, this model was developed, trained, and based off publicly available data, meaning no private patient information was used. However, models trained on medical images, still carry privacy risks. For example, membership inference attacks can allow someone to determine whether a specific data point was included in the training set (Mithila, 2026). To reduce these risks, future medical imaging models should consider using privacy-preserving methods such as differential privacy.

Limitations and Future Improvements:

There are several limitations within this project. Firstly, the model was only trained for 10 epochs, due to computational limits, which likely prevented it from reaching its potential to fully learn and reach its best possible accuracy. Having more training time and additional epochs, the model would likely perform better and more consistently. Secondly, the validation set contains only 1,748 images across six classes, which is relatively small and led to unstable validation results during training. For example, having a larger and more balanced validation set of around 2500 and 3000 images would provide more reliable and stable feedback. Thirdly, the dataset does not include demographic data, making it impossible to determine if the model performs equitably across different population groups. This is critical for the model to perform without any bias. Future improvements would include training for more epochs, expanding the validation set, and incorporating demographic metadata (e.g., age, sex, and geographic background) to evaluate model performance across different population groups; to ensure fairness and equity in predictions.

References

Asak, M. (2024). ChestX6: Multi-class X-ray dataset [Dataset]. Kaggle. https://www.kaggle.com/datasets/mohamedasak/chest-x-ray-6-classes-dataset

Cross, J. L., Choma, M. A., & Onofrey, J. A. (2024). Bias in medical AI: Implications for clinical decision-making. PLOS Digital Health, 3(11), e0000651. https://doi.org/10.1371/journal.pdig.0000651

Driessen, R., Eufracio Plata, A. E., van Assen, M., Roos, C. J., & Nathoe, H. (2024). Enhancement of fairness in AI for chest X-ray classification. JAMIA Open, 7(2). https://doi.org/10.1093/jamiaopen/ooae041

Google. (2018). Artificial intelligence at Google: Our principles. https://ai.google/principles

Hasan, M. M., Islam, M. U., Shiqhre, M. I., & Alam, M. A. (2023). The effectiveness of deep learning vs. traditional methods for lung disease diagnosis using chest X-ray images: A systematic review. Biomedical Signal Processing and Control, 87, 105638. https://doi.org/10.1016/j.bspc.2023.105638

Hasan, M., Ahmed, K., Uddin, M. A., Ali, M. H., & Bui, F. M. (2024). Explainable lung disease classification from chest X-ray images utilizing deep learning and XAI. arXiv. https://arxiv.org/abs/2404.11428

Mithila, M. S. (2026). AI ethics, fairness, and privacy [Lecture notes]. MBAI 5310G: AI Programming. Ontario Tech University.

Obermeyer, Z., Sharchilev, B., Sundin, P., Svahn, J., & Ghassemi, M. (2024). The limits of fair medical imaging AI in real-world generalization. Nature Medicine, 30, 1859–1871. https://doi.org/10.1038/s41591-024-03113-4

Ullah, Z., Usman, M., Latif, S., & Gwak, J. (2023). A deep convolutional neural network model for lung disease detection using chest X-ray imaging. PMC. https://pmc.ncbi.nlm.nih.gov/articles/PMC12213050/

Wasay, A. (2026). Chest_X-Ray_Classifer [GitHub repository]. GitHub. https://github.com/waz-abd/Chest_X-Ray_Classifer